

Auteur: Luca Letroye
 Studierichting: Informatica
 Oefeningenreeks 2E nr. 4.8

$$\begin{aligned}
 & \lim_{x \rightarrow \infty} \left(\frac{ax+b}{ax+c} \right)^{dx+f} \\
 & \lim_{x \rightarrow \infty} \left(\frac{ax+c+b-c}{ax+c} \right)^{dx+f} \\
 & = \lim_{x \rightarrow \infty} \left(1 + \frac{b-c}{ax+c} \right)^{dx+f} \\
 & = \lim_{x \rightarrow \infty} \left(\left(1 + \frac{1}{\frac{ax+c}{b-c}} \right)^{\frac{ax+c}{b-c}} \right)^{\frac{b-c}{ax+c} \cdot (dx+f)} \\
 & = e^{\lim_{x \rightarrow \infty} \frac{d(b-c)}{a}} \\
 & = e^{\frac{bd-cd}{a}}
 \end{aligned}$$

References